

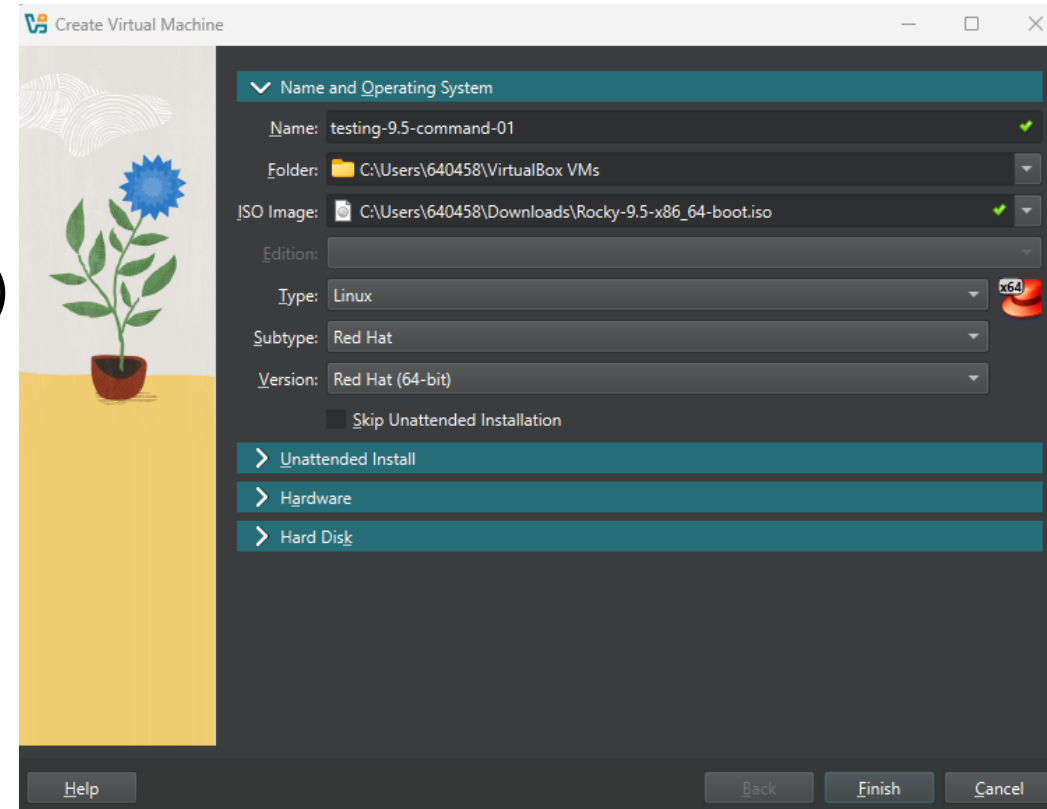
Setting UP TAK Server Virtual Machine

Download/Setup our VM

- Go to the following link:
<https://www.virtualbox.org/wiki/Downloads>
- Under “VirtualBox 7.0.20 platform packages” select your operating system host (Windows hosts)
- Once it is done downloading open the file and following the set-up instructions

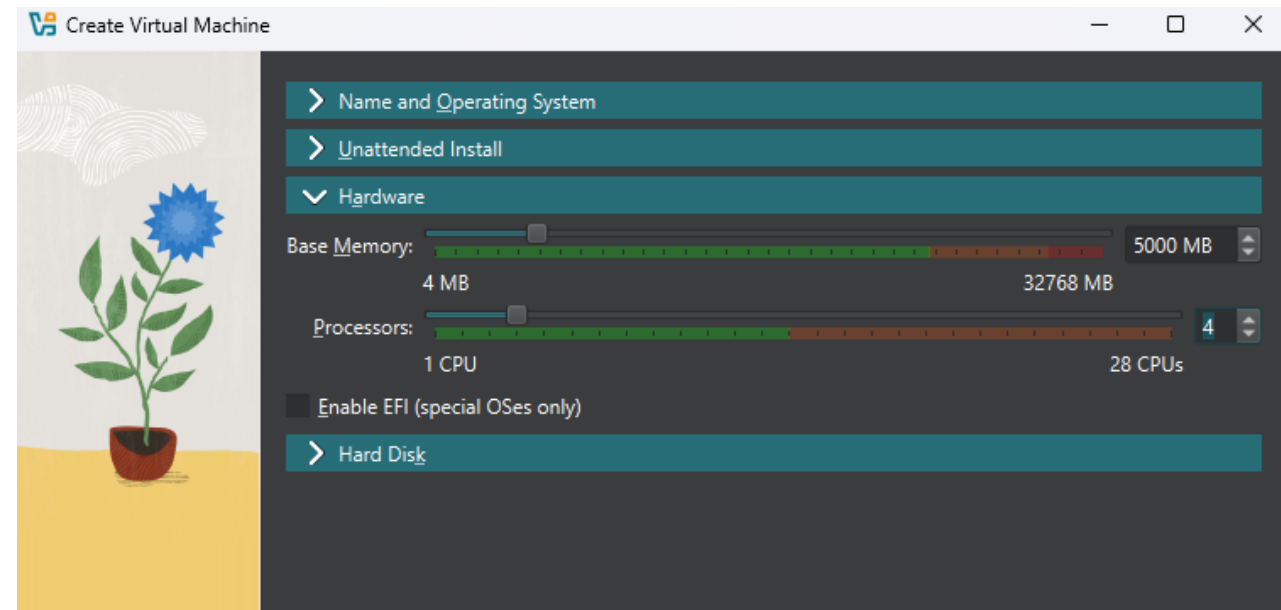
Create Our TAK Server VM

- With the VM open select “New”
- Fill in the following fields:
 - Name: “Name your TAK Server” (TAK Server)
 - ISO Image: Rocky-9.5-x86_64-boot.iso
 - You should be able to select the drop down and see the Rocky boot iso.
 - Type: Linux
 - Version: Red Hat (64-bit)
- Open the Hardware section



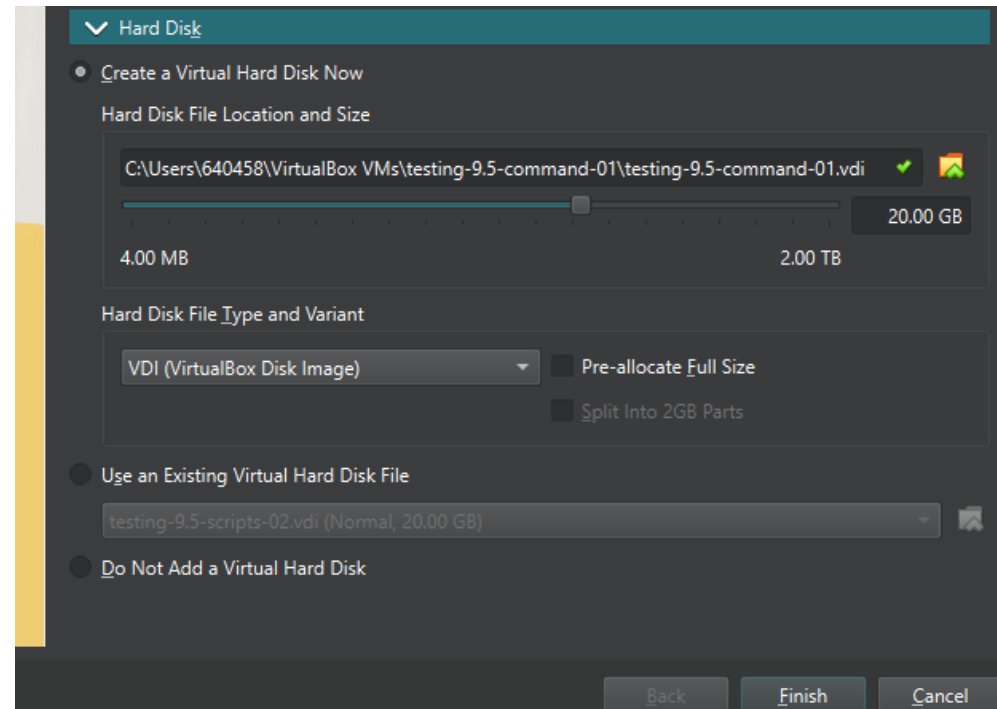
Creating Our TAK Server VM – Hardware

- Base Memory: 5000 MB
 - We want to have around 5 Gb of memory, but it all depends on your machine.
 - You can always add more.
- Processors: 4 CPUs
 - Similarly to the memory this all depends on your machine, you can always add more.
- Open the Hard Disk section



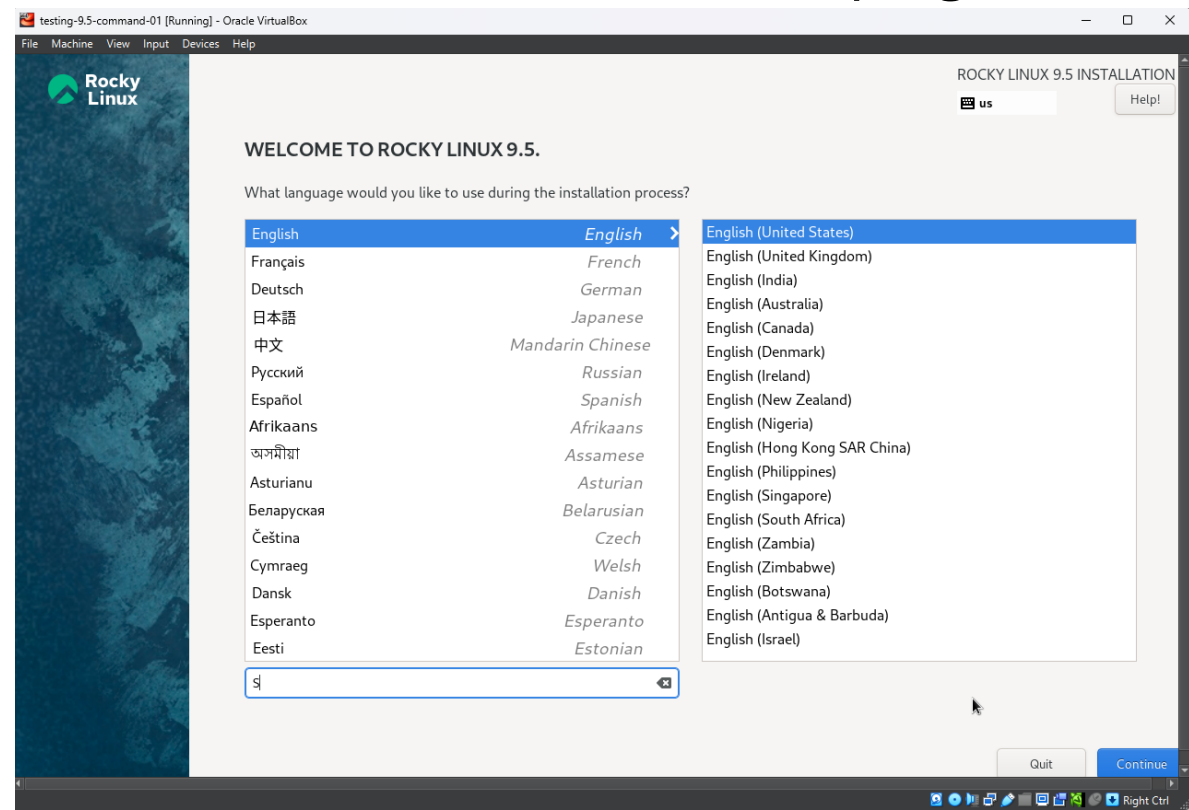
Creating Our TAK Server VM – Virtual Hard Disk

- Select “Create a Virtual Hard Disk Now” if it’s not already.
- Set the “Disk Size:” to a minimum of 20 GB
 - Again, this can be increased to whatever desired size, depending on your machine's specs.
- Select “Finish”



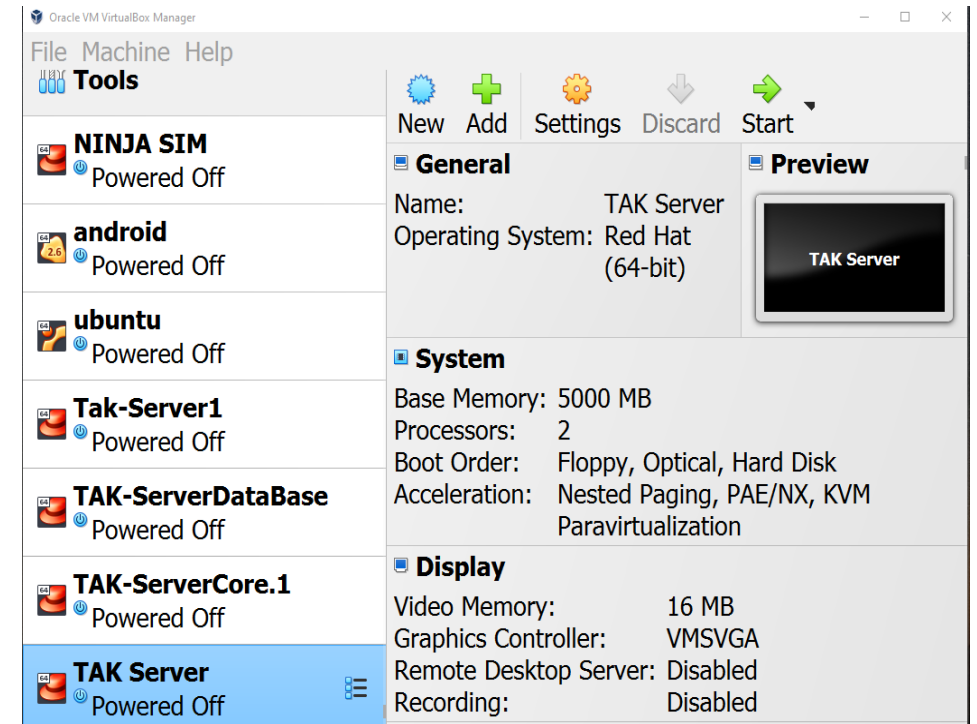
Creating Our TAK Server VM

- The VM will automatically boot when “Finish” is selected, we need to let it fully boot then close it to update the network settings.
 - Once you are at the Welcome screen, selected exit in the top right and power down the VM.
 - Note: To use your mouse cursor outside of your VM select your right “ctrl” key



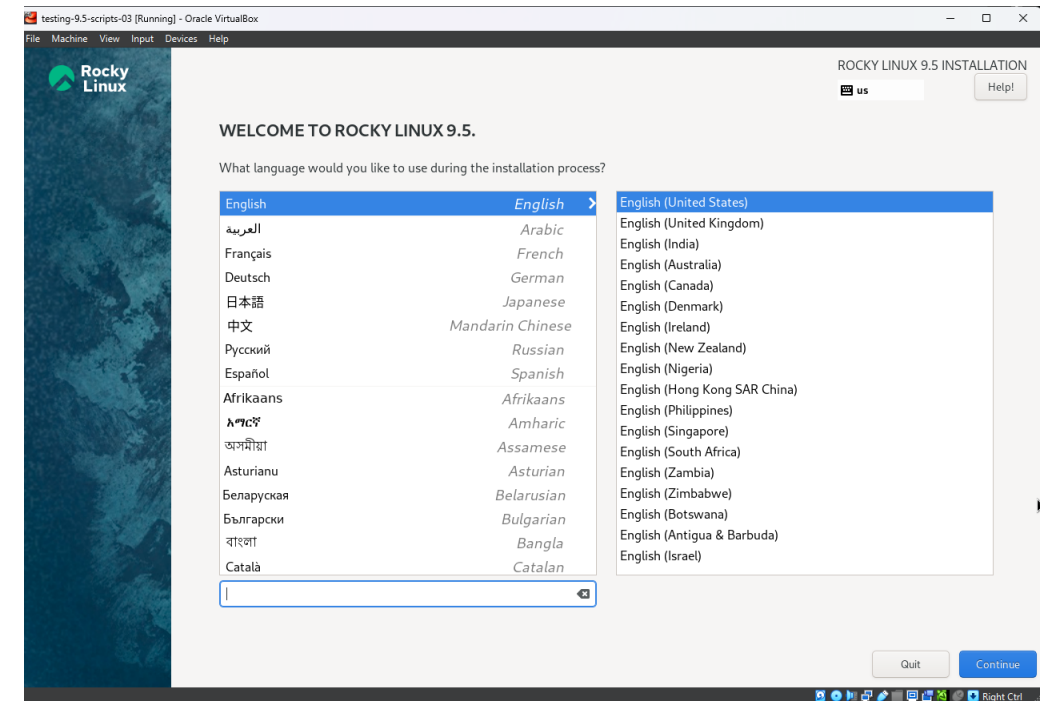
Creating a Bridge Network

- Navigate to the VM Virtual Box.
- With your TAK Server highlighted select “Settings”
- From the settings do the following:
 - Select Network
 - Select Adapter 2
 - Enable: Enable Network Adapter
 - Set Attached to: Bridge Adapter
 - Select Advanced
 - Set Promiscuous Mode: Allow All
 - Disable Adapter 1
 - Select “OK”



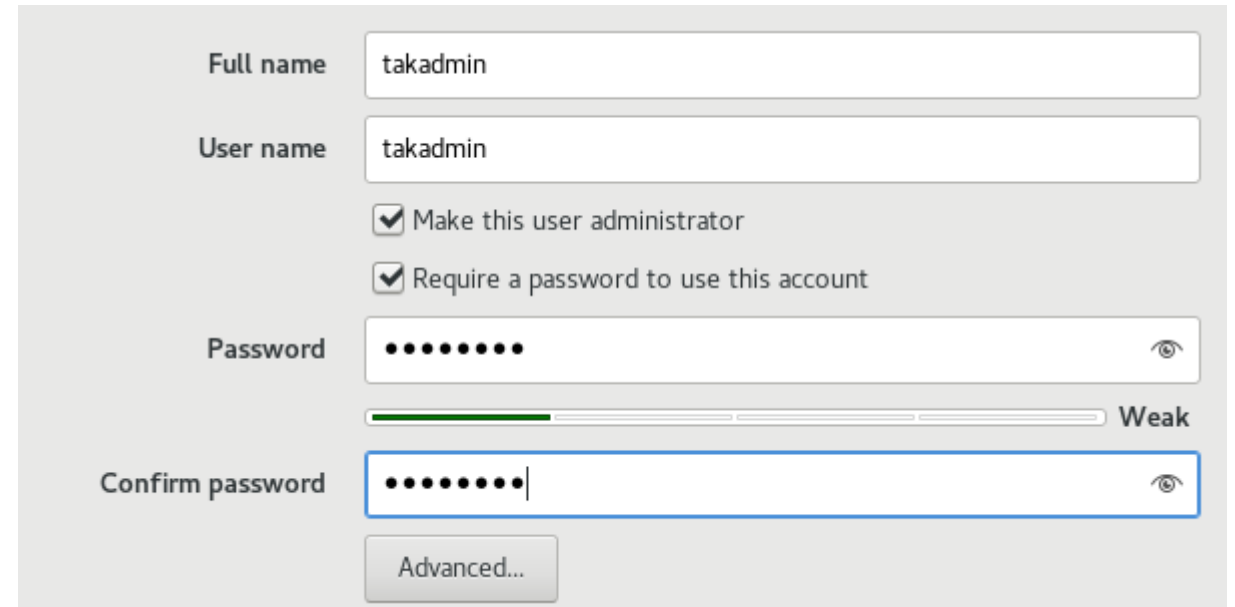
Configure Rocky Linux

- Select start or double click your server to start it.
- Once at the “Welcome to Rocky Linux 9.5” screen walk through the setup guide.
 - Language: Choose your preferred language
 - Continue



Rocky Linux Installation Summary – User Settings

- Once at the “Installation Summary” page complete the “User Settings”
- Select “Root Password”
 - Provide a root password, something not overly complicated.
 - If the password is simple, it is ok you will need to select the “Done” button twice.
 - Root Password: atakatak
 - Confirm the password by typing it again.
 - Select “Done” at the top right.
- Select “Create User”
 - Full name: provide a name (takadmin)
 - Username: (this will automatically be the same as the Full name if you wish to change it you may, but not necessary)
 - IMPORTANT: Make sure the “Make this user administrator” box is enabled
 - Password: provide a password and confirm it, for simplicity I will use the same one as the root admin (atakatak)
 - Select “Done”

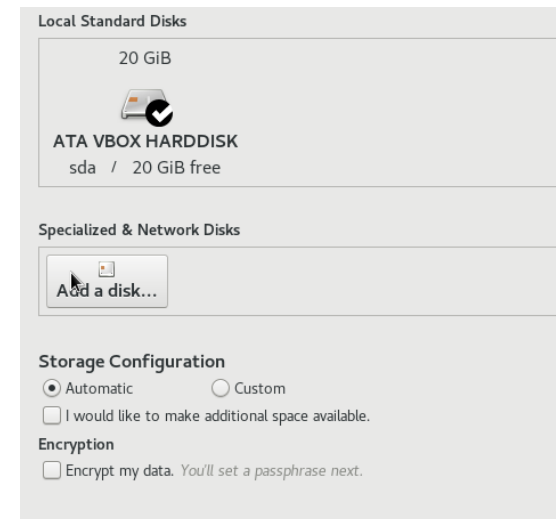


The screenshot shows the 'User Settings' window in the Rocky Linux installer. It contains the following fields and options:

- Full name:** takadmin
- User name:** takadmin
- ☒ Make this user administrator
- ☒ Require a password to use this account
- Password:** A field with 8 dots. Below it is a strength indicator bar that is mostly green, with the word 'Weak' on the right.
- Confirm password:** A field with 8 dots.
- Advanced...** button at the bottom.

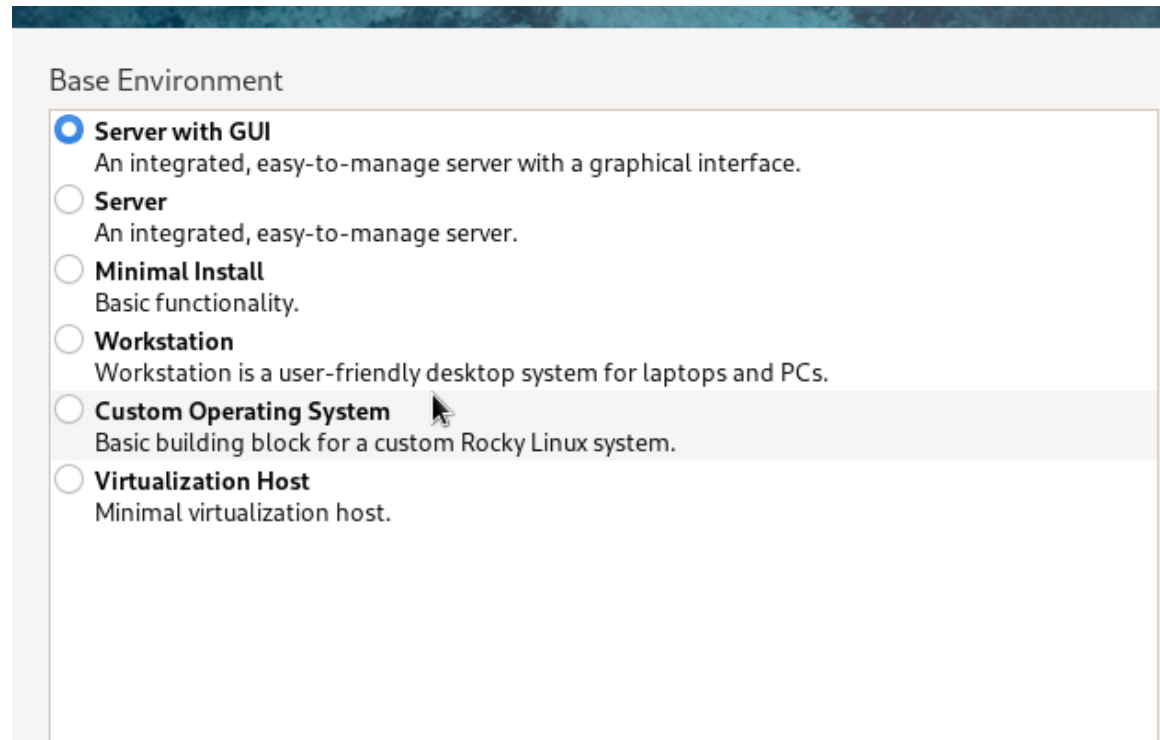
Rocky Linux Installation Summary – System

- Select “Network & Host Name” under the “SYSTEM” section.
 - For Ethernet (enp0s8), toggle the switch on the right side of the screen from to “on” if not already.
 - The switch for Ethernet (enp0s3) should be off, if available.
 - Select “Done”
- Select “Installation Destination” under the “SYSTEM” section.
 - Confirm that the Local Standard Disks has a check mark over the drive.
 - Confirm that the Storage Configuration section is set to Automatic
 - Select “Done”



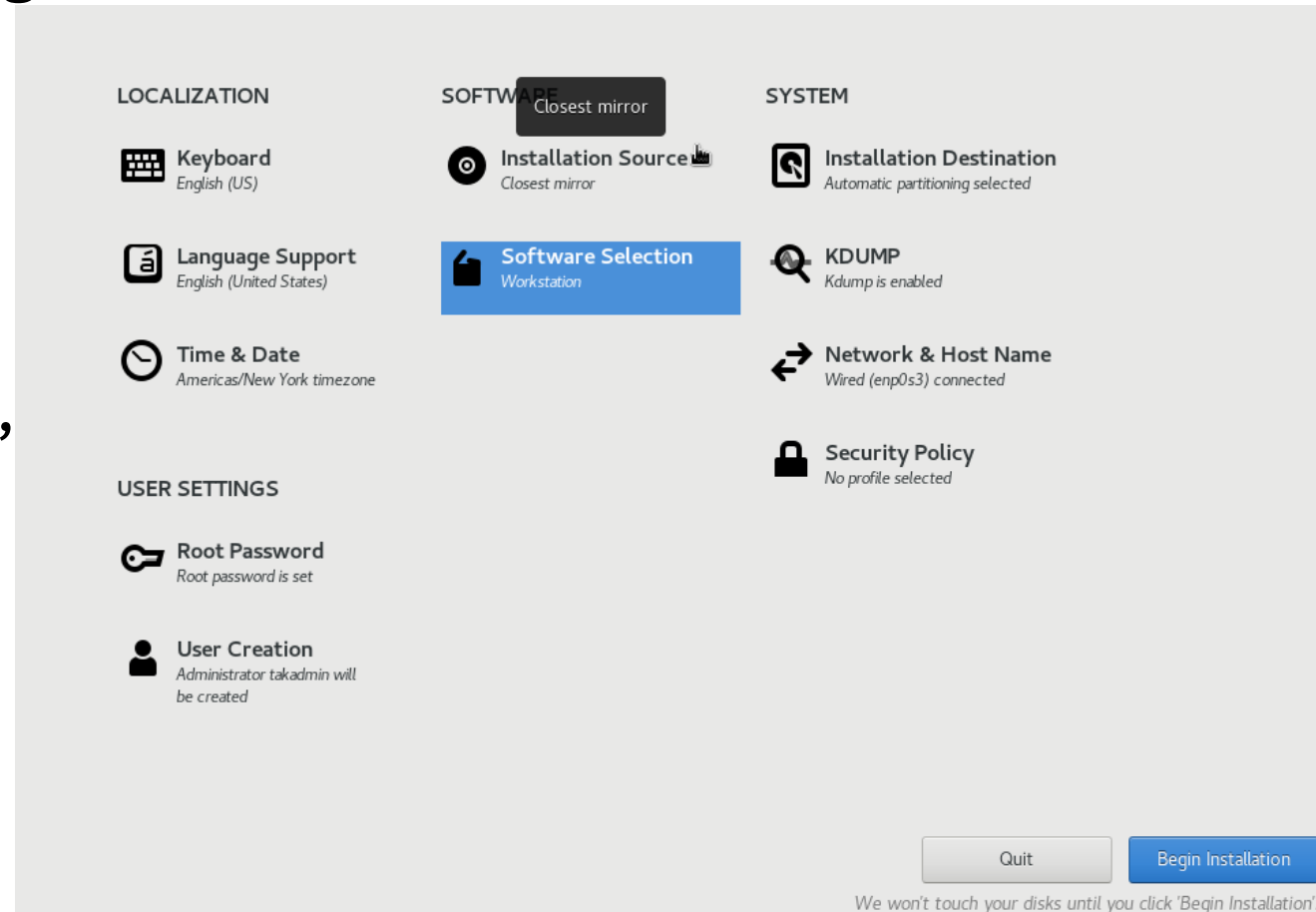
Rocky Linux Installation Summary – Software

- “Software Selection” should automatically be set to “Server with GUI”.
- If it is not, select “Software Selection” In the “Base Environment” section select the “Server with GUI” option.
- Select “Done”



Rocky Linux Installation Summary – Begin Installation

- The “Installation Summary” page should look as follows:
- If it does, then select: “Begin Installation”
- Note: This will take a while.
- Once the installation is complete the “Reboot System” will now be blue and you can select it.



Configuring Network Settings

- Select your user and login.
- Finish setting up.
- Open the terminal and type the following:
 - **ifconfig**
 - This will display the current network interface configuration information.
- If you do not see a valid network IP address, you will need to shut down your VM and set up the Bridge network again.

```
enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.125 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 2600:1700:31f0:6fb0::43 prefixlen 128 scopeid 0x0<global>
    inet6 2600:1700:31f0:6fb0:a00:27ff:fe2b:a502 prefixlen 64 scopeid 0x0<
global>
```

SSH(Secure Shell) into our TAK Server – Terminal

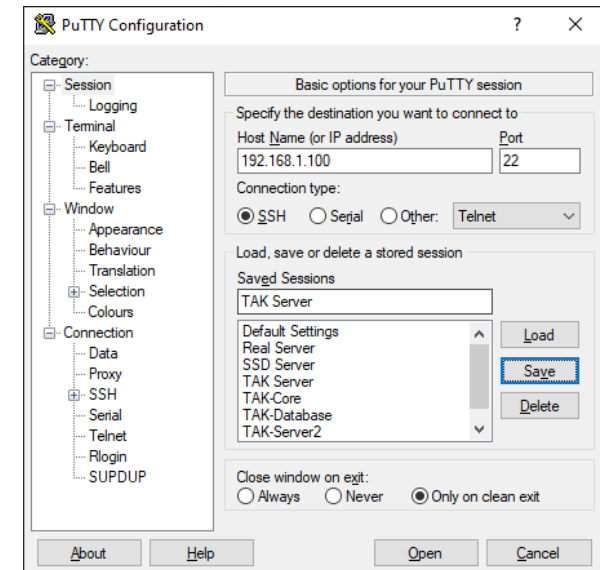
- Local machine Terminal
 - You can ssh into your server from the terminal command line by opening your preferred terminal and typing the following:
 - **ssh <username>@<ip address>**
 - This will ask you to verify you want to connect, type “yes” and press enter.
 - You will then be asked for the password for the user.
 - Note: the text will be invisible so you will not be able to see what you are typing.
 - Once you enter the password hit enter and you will be connected.
 - When you successfully ssh into the server the command line will look like the following:

```
C:\Users\640458>ssh takadmin@192.168.1.125
The authenticity of host '192.168.1.125 (192.168.1.125)' can't be established.
ECDSA key fingerprint is SHA256:jrcXR2DJmw+r1+KByQE9uA+EAU94Qs2WlWRBcoyjh8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.125' (ECDSA) to the list of known hosts.
takadmin@192.168.1.125's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Fri Aug 23 09:49:42 2024
[takadmin@localhost ~]$
```

SSH(Secure Shell) into our TAK Server – PuTTY

- PuTTY
 - PuTTY is a tool that can be leveraged on windows to connect to other clients instead of using the terminal to ssh.
 - It can be downloaded at the following link:
 - <https://www.putty.org/>
 - Once open you will enter the following:
 - Host Name (or IP address): <IP address>
 - Connection Type: SSH
 - Saved Sessions: <Server Name>
 - Select “Save” and the session will be saved for ease of access later.
 - When connecting again later you simply have to select the session you want to connect to then “Load”.
 - Now select “Open”



SSH(Secure Shell) into our TAK Server – PuTTY

- You will be prompted with a “PuTTY Security Alert”
 - Select “Accept”
- You now can login to the client
 - Login as: <username>
 - <username>@<IP address>’s password: <enter your password >
 - The password text will be invisible so you will not be able to see what you are typing.
 - Once done enter your password hit Enter.
 - You should be connected to the server.

Setup TAK Server